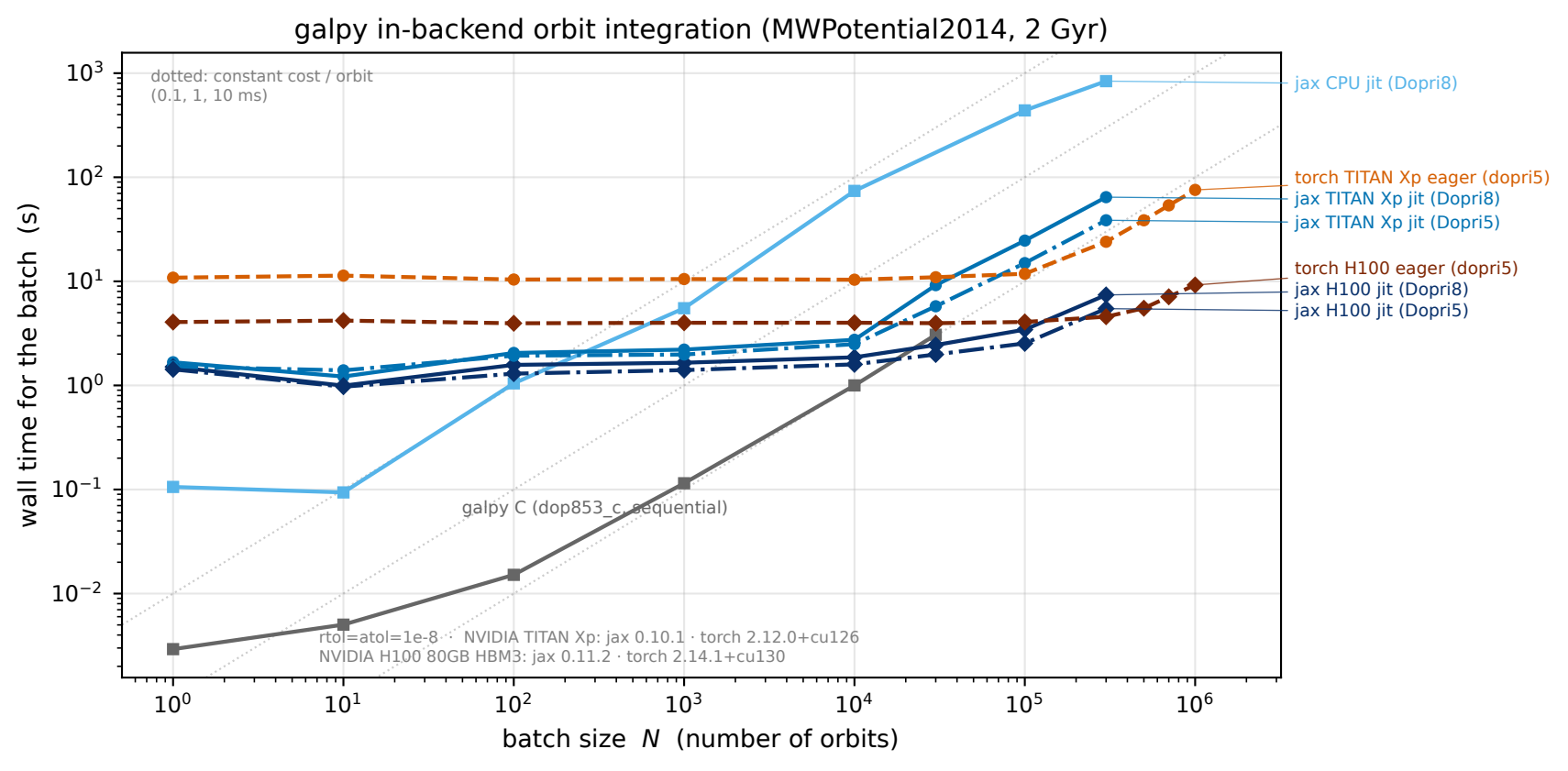
galpy in-backend orbit integration (MWPotential2014, 2 Gyr)
dotted: constant cost / orbit
(0.1, 1, 10 ms)
jax CPU jit (Dopri8)
torch TITAN Xp eager (dopri5)
jax TITAN Xp jit (Dopri8)
jax TITAN Xp jit (Dopri5)
torch H100 eager (dopri5)
jax H100 jit (Dopri8)
jax H100 jit (Dopri5)
galpy C (dop853_c, sequential)
rtol=atol=1e-8 · NVIDIA TITAN Xp: jax 0.10.1 · torch 2.12.0+cu126
NVIDIA H100 80GB HBM3: jax 0.11.2 · torch 2.14.1+cu130
wall time for the batch (s)
batch size N (number of orbits)